# Paleomagnetic and geobarometric study of the Late Cretaceous Mount Lorne stock, southern Yukon Territory[1]

M.J. Harris, D.T.A. Symons, W.H. Blackburn, and C.J.R. Hart

**Abstract**: The ~75 Ma Mount Lorne monzodiorite stock has been studied paleomagnetically to estimate the tectonic motion of the northern Canadian Cordillera since the Late Cretaceous. The stock is one of several paleomagnetic studies currently underway at the University of Windsor Paleomagnetic Laboratory which are funded by the Lithoprobe – Slave northern Cordillera Lithospheric Evolution (SNORCLE) Project. Al-in-hornblende geobarometry and plagioclase-amphibole geothermometry data suggest that the stock has not been tilted since emplacement but does provide evidence that the stock has a normal fault trending north–south through its centre with the east side uplifted relatively by ~1000 m. Paleomagnetic measurements from 12 of 19 granitoid sites yield a well-defined characteristic remanent magnetization direction that is south-southwest and up, and a further six sites yield a direction that is north-northeast and down. Data from a mafic dike yield a negative contact test, suggesting that the dike is coeval with the stock. Combining the 18 granitic site mean characteristic remanent magnetization directions yields a paleopole at 69.1°W, 78.3°N ($d_p$ = 4.1°, $d_m$ = 4.5°) which suggests that the host Stikine Terrane has been translated poleward by 10.5 ± 3.5° (1170 ± 390 km) and rotated clockwise by 57 ± 11° relative to the North American craton between 75 and 50 Ma. Except for the estimate from the nearby coeval Carmacks Group volcanic rocks, the Mount Lorne estimate is consistent with all other paleomagnetic results within the Stikine Terrane. These latter estimates are also consistent with plate-tectonic models that suggest the Pacific oceanic plates had an increased velocity northwards during the Late Cretaceous.

**Résumé** : Une étude paléomagnétique du petit massif intrusif de Mount Lorne, formé de monzodiorite et âgé de ~75 Ma, a été réalisée pour estimer le déplacement tectonique de la partie septentrionale de la Cordillère canadienne depuis le Crétacé tardif. L'étude de ce massif intrusif fait partie de plusieurs autres travaux paléomagnétiques en cours au Laboratoire de paléomagnétisme de l'Université de Windsor, et subventionnés par le projet Lithoprobe–SNORCLE. Les données de géobarométrie de l'Al dans la hornblende et de la géothermométrie de plagioclase-amphibole suggèrent que le massif intrusif n'a pas été dérangé depuis sa mise en place, mais, d'autre part elles révèlent l'existence d'une faille normale de direction nord-sud, qui passe par le centre du massif intrusif et dont le compartiment oriental fut soulevé relativement de ~1000 m. Les mesures paléomagnétiques de 12 sur 19 sites de granitoïde fournissent une direction d'aimantation rémanente caractéristique bien définie sud-sud-ouest vers le haut, tandis que six sites supplémentaires fournissent une direction nord-nord-est vers le bas. Un dyke mafique a donné des résultats qui indiquent que le test du contact est négatif, ce qui suggère que le dyke et le massif intrusif sont contemporains. Une combinaison des directions rémanentes d'aimantation caractéristiques moyennes de 18 sites granitiques fournit le paléopôle de 69,1°W, 78,3°N ($d_p$ = 4,1°, $d_m$ = 4,5°), suggérant que le terrane de Stikine hôte a été déplacé vers le pôle de 10,5 ± 3,5° (1170 ± 390 km) et qu'il a subi une rotation dans le sens des aiguilles d'une montre de 57 ± 11° par rapport au craton Nord américain, entre 75 et 50 Ma. À l'exception des estimations faites pour les roches volcaniques contemporaines avoisinantes du Groupe de Carmacks, les données ontenues pour le Mount Lorne sont compatibles avec les autres résultats paléomagnétiques de mesures effectuées sur des roches de l'intérieur du terrane de Stikine. Plus est, ces dernières estimations sont compatibles avec les modèles de plaque tectonique qui suggèrent une accélération du mouvement vers le nord des plaques océaniques du Pacifique durant le Crétacé tardif.

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## Introduction

The ~75 Ma granodioritic Mount Lorne stock intrudes strata of the Whitehorse Trough in the Stikine Terrane of the northern Intermontane Belt to the southeast of Whitehorse, Yukon Territory (Fig. 1). It is one of several targets for Lithoprobe – Slave northern Cordillera Lithospheric Evolution (SNORCLE) paleomagnetic studies being done at the University of Windsor to understand better the geotectonic history of the northern Cordillera.

Results from two previous studies in the southern Yukon suggest that the area has undergone 11–14° of poleward translation and ~60° of clockwise rotation since ~110 Ma (Harris et al. 1996, 1997). These results are also consistent with those from several contemporaneous bodies in central and southern British Columbia (Monger and Irving 1980; Symons 1983; Irving and Thorkelson 1990; Irving et al. 1995). In contrast, the ~70 Ma Carmacks Group volcanic rocks, the only other unit studied paleomagnetically in southern Yukon, gave a much higher translation estimate (Marquis and Globerman 1988; Johnston et al. 1996; Wynne et al. 1998) that is inconsistent with the other six studies.

Here we report paleomagnetic data from the Mount Lorne stock and the resulting estimate of northward displacement of the Stikine Terrane relative to the North American craton for Late Cretaceous time. Further, geothermobarometric data on hornblende and plagioclase crystals are presented to measure the postmagnetization tilting of the stock to correct the paleomagnetic data as necessary.

## Geology

The Mount Lorne stock is exposed ~30 km southeast of the city of Whitehorse, Yukon. The stock has an areal exposure of ~30 $km^2$ (Fig. 2), although an aeromagnetic survey suggests the stock is ~10% larger below the surface. The aeromagnetic signature matches the known outcrop pattern and both are circular. The stock was first mapped geologically by Wheeler (1961), with a recent revision by C.J.R. Hart (unpublished data).

The stock intrudes three stratigraphic units that range in age from Upper Triassic to Upper Cretaceous (Fig. 2). Rocks from the Upper Triassic Lewes River Group consist of a clastic unit comprising wacke, arkose, and sandstone and a limestone unit. Rocks of the Early to Middle Jurassic Laberge Group consist of wacke, arkose, and sandstone. The Upper Cretaceous volcanic rocks that crop out to the southwest generally consist of andesitic flows and are correlated with the Wheaton River volcanic suite (Hart and Radloff 1990; Hart 1995). Host-rock bedding attitudes are variable around the stock because the two older groups were deformed in the Middle Jurassic (Hart and Radloff 1990). The volcanic rocks dip to the east with moderate to shallow angles. They remain preserved, however, as they were downdropped along a north-trending fault with west-side-down normal motion. The fault trends into the pluton but, since similar rock units are juxtaposed, the extent of the fault within the pluton is uncertain (Fig. 2).

The stock is composed of massive medium-grained biotite–hornblende granodiorite and monzogranite. Micropetrography shows that most samples contain 25–50% euhedral plagioclase ($An_{15}$–$An_{30}$), 14–26% subhedral to interstitial potassium feldspar, and 17–40% anhedral to interstitial quartz. Mafic minerals include up to 14% euhedral to anhedral hornblende and up to 11% biotite. Accessory minerals include up to a few percent of magnetite, titanite, and apatite. Hornblende crystals commonly have clinopyroxene cores. Alteration is minimal, with only minor selective sericitization of plagioclase and chloritization of amphiboles.

A K–Ar date on hornblende gave an age of 75.3 ± 2.8 Ma (Morrison et al. 1979). The calculated $^{87}Sr/^{86}Sr$ initial ratio from the same sample was 0.7044 ± 0.0002. The Wheaton River volcanic rocks in the study area yielded a whole rock K–Ar date of 80.8 ± 2.8 Ma (Hart 1995), defining a maximum age for the Mount Lorne stock.

## Geobarometry

The Al-in-hornblende geobarometer and amphibole–plagioclase geothermometer were used to estimate the pressure and temperature of crystallization. The theory and procedures have been described in detail elsewhere (Blundy and Holland 1990; Anderson and Smith 1995; Harris et al. 1996, 1997; Harris 1998).

Eight sites from six localities were chosen to represent the stock spatially. Chemical analyses on the hornblende and plagioclase pairs were performed on the JEOL JXA-8600 Superprobe at the University of Western Ontario. Operating conditions were set at a probe current of 10 nA, an accelerating voltage of 15 kV, a focused beam of 2 μm for amphibole and 5 μm for plagioclase, and a 20 s accumulation time. The standards used for calibration of amphibole are as follows: Si, Cr augite, USNM 164905; Al, anorthite; Cr, chromite, USNM 117075; Fe, fayalite; Mn, rhodonite; Mg, olivine, USNM 111312; Ca, Wakefield diopside; Na, Amelia albite; K, orthoclase; F, LiF; Cl, Tugtupite; O, by difference. The standards used for calibration of plagioclase are as follows: Si, Amelia albite; Al, anorthite; Fe, fayalite; Ca, anorthite; Na, Amelia albite; K, orthoclase; O, by difference.

Representative amphibole analyses, site-averaged compositions, and pressure–temperature–depth (*P–T–d*) data are given in Tables 1 and 2 and Figs. 3 and 4. Amphibole structural formulae are calculated on the basis of 13 cations, a method that best predicts site occupancies and most reasonable estimates of $Fe^{3+}$ and $Fe^{2+}$ (Cosca et al. 1991; Harris et al. 1997). The plagioclase structural formulae are calculated on the basis of 32 oxygens. The amphiboles exhibit a small compositional range (Table 1; Fig. 3) and most are classified as magnesiohornblendes. The amphiboles from site ML15 are more iron rich and are classified as ferrohornblendes. Four of the 14 amphiboles have lower aluminum and higher silica concentrations (samples 8.2, 16.1, 16.2, 19.2) (Fig. 3*c*). Petrographic observations show that these sections contain altered plagioclase and amphibole, and thus these results are not considered in the averages.

Preliminary pressures range from 133 to 222 MPa, and temperatures range from 660 to 684°C (Table 2). Final temperature-corrected pressures (Anderson and Smith 1995) range from 140 to 225 MPa, and the corresponding depths range from 5.03 to 8.11 km (Table 2; Fig. 4). The $Fe^{3+}/(Fe^{3+} + Fe^{2+})$ ratio is generally above 0.20, and the

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**Fig. 1.** Morphogeological belts of the Canadian Cordillera. AX, Axelgold intrusions; CK, Carmacks Group volcanic rocks; EN, Endako intrusions; ML, Mount Lorne stock; MM, Mount McIntyre pluton; SB, Spences Bridge volcanic rocks; WP, Whitehorse batholith. After Harris et al. (1997).

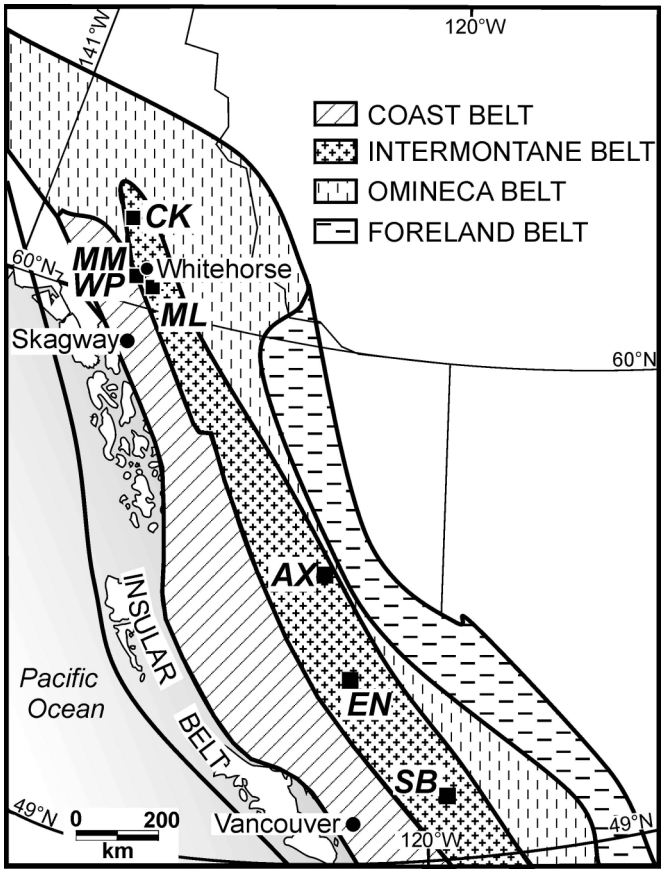
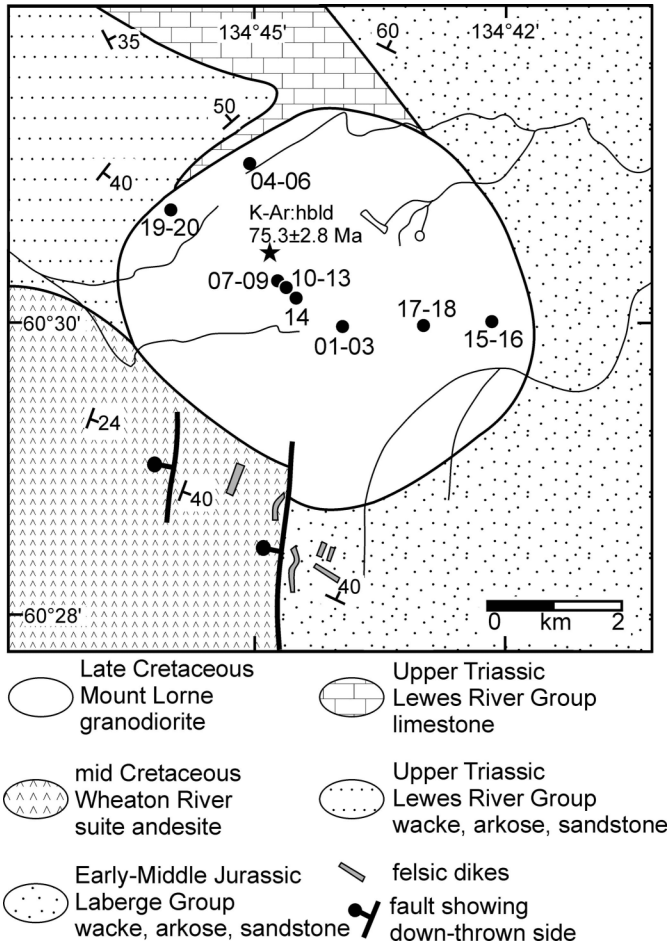



**Fig. 2.** Geology of the Mount Lorne stock after C.J.R. Hart (unpublished data) with sampling sites.




$(Fe^{2+} + Fe^{3+})/(Fe^{2+} + Fe^{3+} + Mg)$ ratio is between 0.34 and 0.57 (Table 1), thus satisfying the requirements presented by Anderson and Smith (1995). The analytical standard deviations on the pressure averages per site (Table 2) are minor and all less than the error on the calibration of the geobarometer (±60 MPa; Anderson and Smith 1995).

## Paleomagnetism

Twenty sites were sampled across the stock to represent it both petrographically and spatially (Fig. 2). At least five cores per site were drilled and oriented in situ using a solar compass. A 0.5 m wide mafic dike was sampled as site ML07, with site ML08 in the host rock a few metres away to provide a paleomagnetic contact test. The other 18 sites were within the massive stock.

All paleomagnetic measurements were done at Windsor using an automated CTF DRM-420 cryogenic magnetometer, a Sapphire Instruments SI-4 alternating field (AF) demagnetizer, a Sapphire Instruments SI-5 spinner magnetometer, a Sapphire Instruments SI-6 pulse magnetizer, and a Magnetic Measurements MMTD-1 thermal demagnetizer. All the instruments are in a magnetically shielded room which has an ambient field of 0.2% of the Earth's magnetic field intensity.

Specimen characteristic remanent magnetization (ChRM) directions were calculated using either the end-point method of Kirschvink (1980) or the remagnetization circle method of Halls (1978). The latter method is used for specimens in which the ChRM direction is not fully isolated as an end-point because of too slow removal of the viscous remanence (VRM) or because of the acquisition of an anhysteretic remanent magnetization (ARM) that partially masks the ChRM. Specimen ChRM directions were accepted for statistical analysis if the best-fit stable direction had a maximum angular deviation of <15° over three or more consecutive steps, with most having much less deviation. Site and unit mean directions were then calculated using the statistical method of Fisher (1953), where directions calculated from the above two methods each carried unit weight.

The natural remanent magnetization (NRM) was measured for each of the 238 specimens. Four pilot specimens having average direction and intensity were selected from each site, two for detailed AF demagnetization and two for detailed thermal step demagnetization. Thermal decay curves from the pilot specimens showed that the dominant magnetic mineral is titanomagnetite. In some cases, however, the magnetite is likely more pure because several specimens have very well defined "knees" with high unblocking temperatures (Fig. 5). A few specimens exhibit intensity increases after the first step or two, before demagnetizing in

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**Table 1.** Representative amphibole analyses from the border zone of the Mount Lorne stock.

| Site: | ML03 | ML05 | ML08 | ML15 | ML16 | ML17 | ML19 | ML20 |
|---|---|---|---|---|---|---|---|---|
| Sample No.: | 03.1c | 05.2c | 08.1c | 15.1a | 16.1a | 17.1b | 19.1c | 20.1b |
| $SiO_2$ (wt.%) | 47.70 | 48.30 | 49.00 | 46.66 | 50.79 | 48.35 | 47.98 | 48.77 |
| $TiO_2$ | 1.06 | 1.24 | 1.08 | 0.99 | 0.46 | 1.14 | 0.97 | 0.97 |
| $Al_2O_3$ | 6.32 | 5.51 | 5.58 | 5.91 | 3.37 | 5.93 | 5.96 | 5.96 |
| $Fe_2O_3$* | 4.15 | 3.38 | 5.20 | 4.61 | 3.64 | 4.38 | 3.82 | 5.65 |
| $Cr_2O_3$ | 0.01 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 | 0.02 | 0.00 |
| FeO | 10.96 | 9.96 | 9.37 | 17.24 | 15.04 | 10.74 | 12.12 | 8.88 |
| MnO | 0.70 | 0.45 | 0.63 | 0.85 | 0.64 | 0.62 | 0.32 | 0.48 |
| MgO | 13.04 | 14.47 | 14.13 | 8.70 | 11.33 | 13.42 | 12.82 | 14.03 |
| CaO | 11.56 | 12.06 | 11.73 | 11.19 | 11.43 | 11.68 | 11.90 | 11.50 |
| $Na_2O$ | 1.43 | 1.18 | 1.12 | 1.11 | 0.68 | 1.27 | 1.09 | 1.06 |
| $K_2O$ | 0.53 | 0.46 | 0.49 | 0.66 | 0.28 | 0.57 | 0.51 | 0.58 |
| $H_2O$* | 1.93 | 1.93 | 1.98 | 1.77 | 1.89 | 1.93 | 1.91 | 1.98 |
| F | 0.20 | 0.21 | 0.14 | 0.40 | 0.27 | 0.22 | 0.22 | 0.11 |
| Cl | 0.05 | 0.05 | 0.08 | 0.09 | 0.04 | 0.06 | 0.07 | 0.12 |
| F,Cl = O | 0.09 | 0.10 | 0.08 | 0.19 | 0.12 | 0.10 | 0.11 | 0.07 |
| Total | 99.54 | 99.10 | 100.46 | 99.99 | 99.74 | 100.22 | 99.61 | 100.02 |
| Si | 7.025 | 7.091 | 7.099 | 7.058 | 7.516 | 7.062 | 7.077 | 7.083 |
| $^{[4]}Al$ | 0.975 | 0.909 | 0.901 | 0.942 | 0.484 | 0.938 | 0.923 | 0.917 |
| $^{[6]}Al$ | 0.123 | 0.045 | 0.052 | 0.112 | 0.104 | 0.083 | 0.113 | 0.103 |
| $Fe^{3+}$ | 0.460 | 0.374 | 0.567 | 0.524 | 0.405 | 0.482 | 0.424 | 0.618 |
| $Fe^{2+}$ | 1.350 | 1.222 | 1.135 | 2.180 | 1.861 | 1.311 | 1.495 | 1.078 |
| Ti | 0.117 | 0.137 | 0.118 | 0.113 | 0.051 | 0.125 | 0.108 | 0.106 |
| Mg | 2.862 | 3.166 | 3.051 | 1.961 | 2.499 | 2.921 | 2.818 | 3.037 |
| Mn | 0.087 | 0.056 | 0.077 | 0.109 | 0.080 | 0.077 | 0.040 | 0.059 |
| Cr | 0.001 | 0.000 | 0.000 | 0.001 | 0.000 | 0.000 | 0.002 | 0.000 |
| Ca | 1.824 | 1.897 | 1.821 | 1.814 | 1.812 | 1.828 | 1.881 | 1.789 |
| $^{[M4]}Na$ | 0.176 | 0.103 | 0.179 | 0.186 | 0.188 | 0.172 | 0.119 | 0.211 |
| $^{[A]}Na$ | 0.233 | 0.233 | 0.136 | 0.139 | 0.007 | 0.188 | 0.192 | 0.088 |
| K | 0.099 | 0.086 | 0.091 | 0.127 | 0.053 | 0.106 | 0.096 | 0.107 |
| OH* | 1.894 | 1.890 | 1.916 | 1.786 | 1.864 | 1.884 | 1.880 | 1.920 |
| F | 0.093 | 0.098 | 0.064 | 0.191 | 0.126 | 0.102 | 0.103 | 0.051 |
| Cl | 0.012 | 0.012 | 0.020 | 0.023 | 0.010 | 0.015 | 0.018 | 0.030 |
| $Fe^{3+}$# | 0.25 | 0.23 | 0.33 | 0.19 | 0.18 | 0.27 | 0.22 | 0.36 |
| Fe# | 0.39 | 0.34 | 0.36 | 0.58 | 0.48 | 0.38 | 0.41 | 0.36 |

**Notes**: Structural formulae are normalized to 13 cations exclusive of Ca, Na, and K. $Fe_2O_3$* and $H_2O$* are back-calculated from the structural formula. $Fe^{3+}\# = Fe^{3+}/(Fe^{2+} + Fe^{3+})$; $Fe\# = (Fe^{2+} + Fe^{3+})/(Fe^{2+} + Fe^{3+} + Mg)$.

the magnetite range. This is likely caused by the polarity reversal these specimens underwent during progressive demagnetization. Twelve of the 40 AF pilot specimens were also used for saturation isothermal remanent magnetization (SIRM) testing. SIRM acquisition and decay curves support the thermal decay curve results and further indicate that the magnetite is mostly pseudosingle domain (Fig. 6).

Typical direction and intensity changes are shown in orthogonal step demagnetization plots (Fig. 7) (Zijderveld 1967). Viscous remanence from modern-day overprinting is generally removed by low temperatures of ~200°C (Fig. 7*a*) or low alternating fields of ~30 mT (Figs. 7*b*, 7*d*). Figure 7*c* exemplifies a few specimens that show a low-temperature, steeply down inclination and a ChRM direction that is directed southwest and moderately upwards in the high-temperature range (500–570°C).

Sites ML15 and ML16 have slightly different magnetic characteristics. Their intensities are generally three orders of magnitude lower than the rest of the stock and their magnetic signature is carried by single-domain rather than pseudosingle-domain magnetite (Fig. 6). The orthogonal plots for these sites are often more erratic and do not always reach the origin (Fig. 7*e*).

The specimens from the mafic dike at site ML07 have their ChRM direction carried by pseudosingle-domain magnetite (Fig. 5). They have a single ChRM component that is directed northward and steeply down (Fig. 7*f*).

Site mean ChRM directions were successfully calculated for 18 granitoid sites and the one dike site (Table 3; Fig. 8). Site ML11 had randomly directed specimen directions, so no mean could be calculated for it. Six of the granitoid sites have ChRM directions to the north-northeast and steeply down and the other 12 sites are antiparallel with ChRM directions south-southwest and steeply up (Fig. 8). The two directions are not statistically different at the 95% confidence level and the two can be combined to form one population.

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**Table 2.** Site-averaged compositions of coexisting amphibole and plagioclase edges and thermobarometric data for the Mount Lorne stock.

| Site | Plag Ab# | Amphibole (avg. $Al_{tot}$) | *P* (Sch) (MPa) | *T* (B and H) (°C) | *P* (A and S) (MPa) | Depth (km) |
|---|---|---|---|---|---|---|
| ML03.1 | 72.6 | 1.075±0.057 | 211 | 682 | 205±11 | 7.38 |
| ML03.2 | 77.7 | 1.100±0.104 | 223 | 669 | 227±21 | 8.17 |
| ML05.1 | 73.7 | 0.912±0.006 | 133 | 669 | 136±1 | 4.90 |
| ML05.2 | 83.4 | 0.969±0.030 | 160 | 658 | 169±5 | 6.08 |
| ML08.1 | 76.2 | 0.974±0.032 | 163 | 671 | 165±5 | 5.94 |
| ML15.1 | 73.0 | 0.993±0.078 | 172 | 667 | 176±14 | 6.34 |
| ML17.1 | 79.8 | 1.099±0.078 | 222 | 681 | 218±18 | 7.85 |
| ML19.1 | 74.3 | 1.046±0.031 | 197 | 673 | 198±6 | 7.13 |
| ML20.1 | 69.5 | 1.000±0.017 | 175 | 678 | 173±3 | 6.23 |
| ML20.2 | 72.2 | 0.995±0.012 | 173 | 678 | 170±2 | 6.12 |

**Notes**: Plag Ab# is the atomic ratio [Na/(Na + Ca)] in plagioclase, *P* (Sch) is from Schmidt (1992), *T* (B and H) is from Blundy and Holland (1990), and *P* (A and S) is from Anderson and Smith (1995).

**Fig. 3.** Hornblende compositions of the Mount Lorne stock. (*a*) Simplified calcic-amphibole classification plot after Leake et al. (1997). (*b*, *c*) Amphibole exchange reactions. Pressures shown in (*c*) are based on the Schmidt (1992) calibration. Labelled points in (*c*) are specimen numbers referred to in the text.

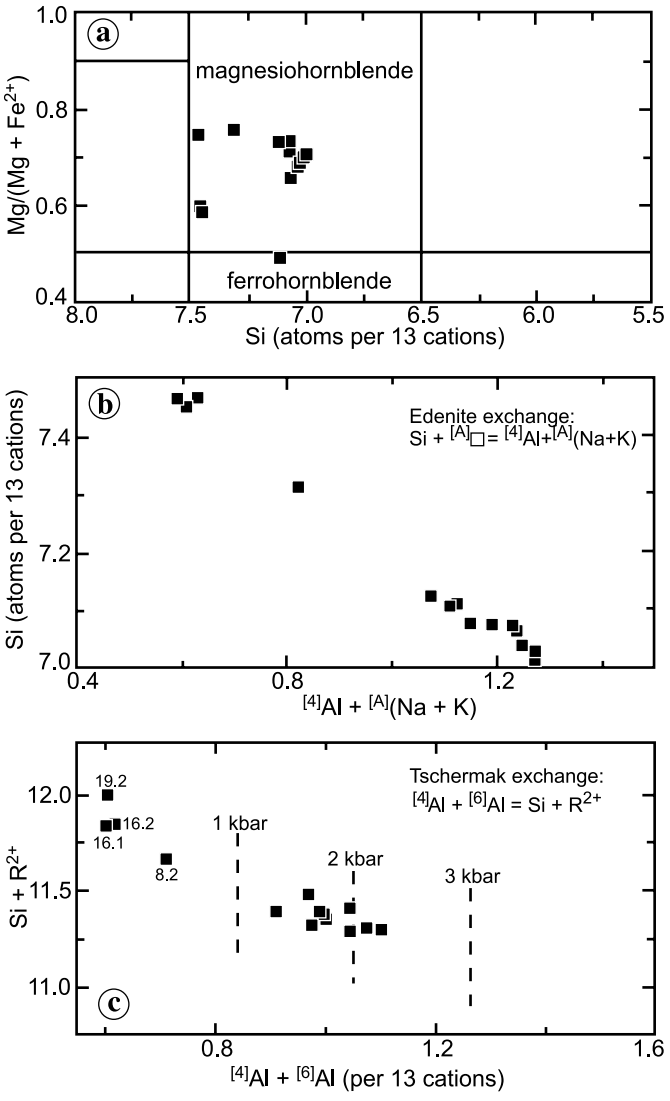
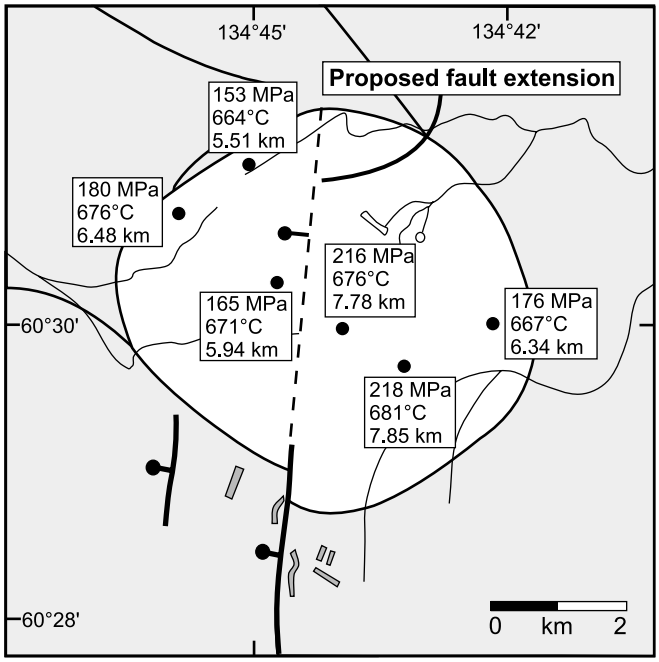



**Fig. 4.** Pressure, temperature, and intrusion depth data from Table 2.




Combining the 18 granitoid site mean directions after reversing the up directions to their antiparallel position gives a unit mean ChRM direction of declination (*D*) = 24.2°, inclination (*I*) = 75.9°, radius of 95% confidence ($\alpha_{95}$) = 2.4°, and precision parameter (*k*) = 201 (Table 3).

The ChRM direction of the mafic dike (site ML07) is north and steeply down, well within the cluster of the granitoid directions. The ChRM directions for the dike and the monzodiorite at that locality (site ML08) are not significantly different at the 95% confidence level (Table 3) and thus constitute a negative contact test. This implies either that the dike is coeval with the stock or that the local area has been remagnetized since the dike intruded the stock. The former is preferred because the ChRM directions from both sites (ML07 and ML08) are not significantly different from the ChRM directions of the rest of the stock (Fig. 8) and be-

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**Fig. 5.** Thermal demagnetization decay curves for the Mount Lorne stock. Solid lines are granitoid specimens and broken lines are mafic dike specimens. P and M, preferred unblocking temperature ranges diagnostic of pyrrhotite and magnetite, respectively. Vertical axis is the ratio of the measured intensity (J) to the NRM intensity ($J_{NRM}$).

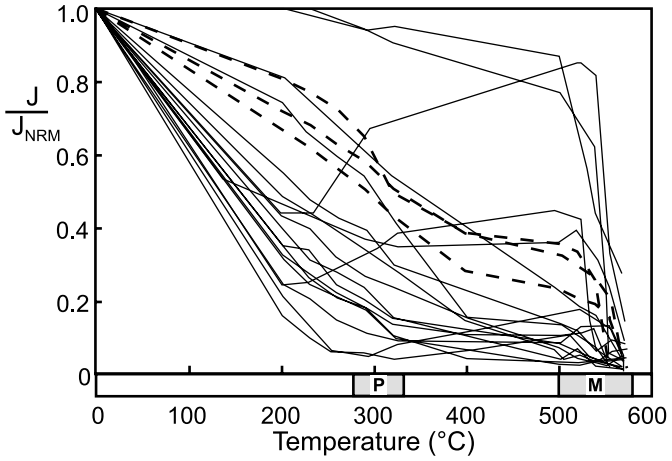
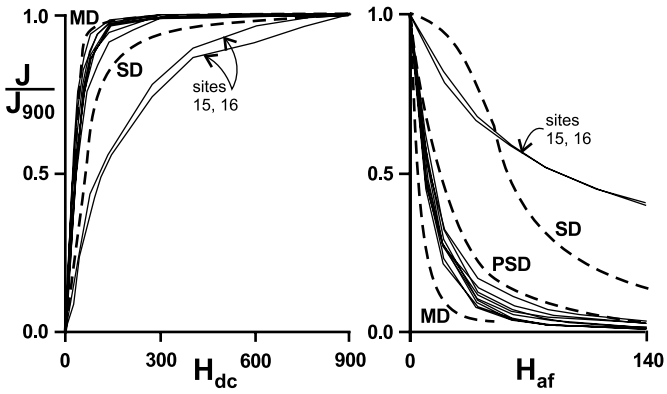



**Fig. 6.** Saturation isothermal remanent magnetization (SIRM) acquisition and decay curves for the Mount Lorne stock. The horizontal axes represent the magnetization acquired ($H_{dc}$) and decayed ($H_{af}$) measured in mT. The vertical axis is the ratio of the measured intensity J to the saturation intensity ($J_{900}$) at 900 mT. The type curves are for single-domain (SD), pseudosingle-domain (PSD), and multidomain (MD) magnetite.




cause there are no post-emplacement events to reheat the stock.

## Discussion

The ~75 Ma Mount Lorne stock postdates the Middle Jurassic deformation of Stikine Terrane strata. Following emplacement of the stock, deformation was localized and largely limited to normal faulting (Hart and Radloff 1990). Igneous textures and mineralogy indicate that the pluton has not been metamorphosed or heated to high enough temperatures to affect either the hornblende chemistry (>600°C) or the magnetic properties (>450°C). Therefore, the reported K–Ar ages, measured pressures, temperatures, and ChRM directions are all considered primary.

### Geobarometry

The maximum topographic relief between the six geobarometry locations in the Mount Lorne stock is ~215 m. This yields a statistically insignificant topographic correction of less than 6 MPa. The geobarometric results have a bimodal geographical distribution, with the three eastern locations having an average emplacement depth of 6.0 km, whereas the three western locations have an average depth of 7.0 km (Fig. 4). Thus, it is proposed that the fault mapped to the south (Fig. 2) (Hart, unpublished data) extends through the stock with a vertical displacement of ~1000 m. The sense and magnitude of this displacement are consistent with those mapped between the volcanic rocks and Laberge Group sediments (Hart, unpublished data). Geobarometric data do not otherwise indicate postmagmatic movement or tilting of the stock on either side of the fault. The circular shape of the pluton and its parallel, circular aeromagnetic signature also suggest that the stock is not tilted.

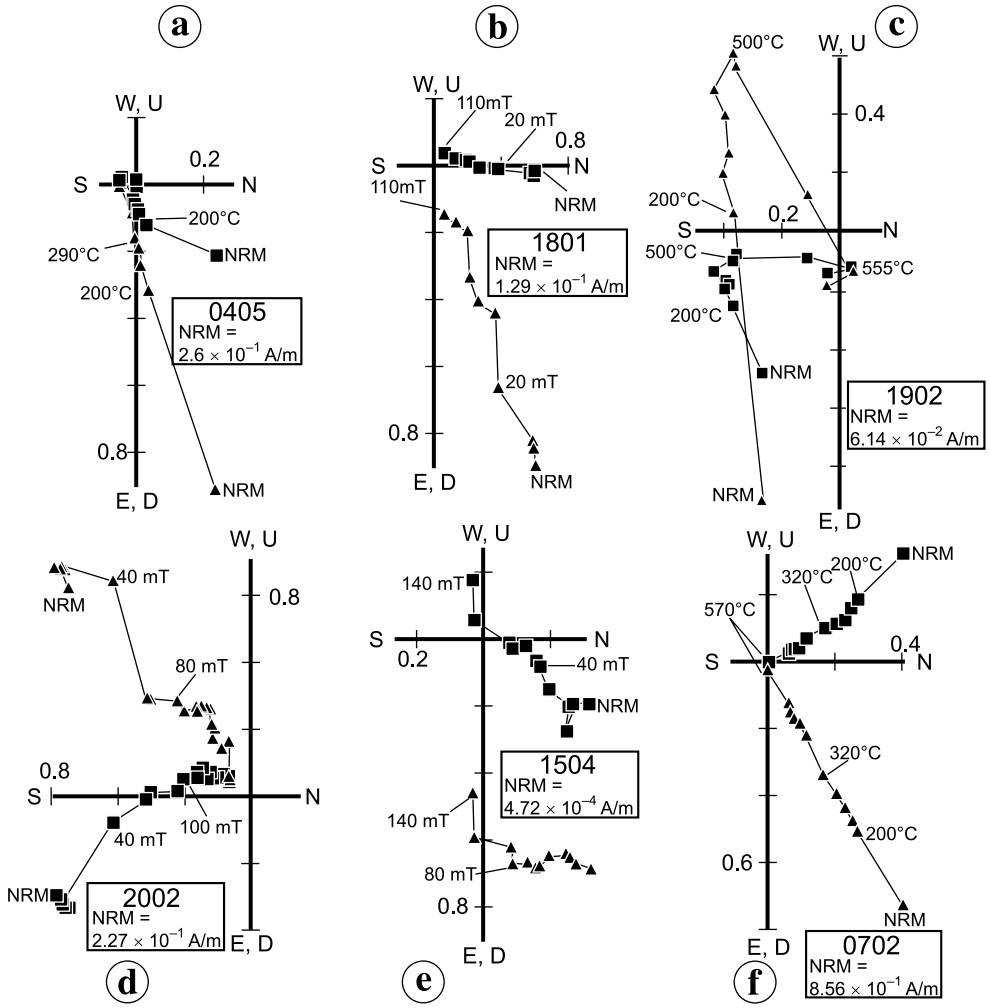
### Tectonics

The mean ChRM direction for the ~75 Ma Mount Lorne stock corresponds to a paleopole of 69.1°W, 78.3°N ($d_p$ = 4.1°, $d_m$ = 4.5°). This pole is compared to the 75 Ma reference pole for the North American craton at 166.0°W, 71.5°N ($A_{95}$ = 3.7°) that was interpolated from the apparent polar wander path (APWP) of Besse and Courtillot (1991). The results indicate that the stock has been translated northward by 10.5 ± 3.5°, equivalent to 1170 ± 390 km, and rotated clockwise by 57 ± 11° relative to the craton (Table 4; Fig. 9). This result is compared with estimates from other rock units that are found in the Stikine Terrane: the 70 Ma Carmacks volcanic rocks in the Yukon; the 103 Ma Axelgold intrusions in central British Columbia; the 104 Ma Spences Bridge volcanic rocks in southern British Columbia; the 109 Ma Mount McIntyre pluton in the Yukon; the ~112 Ma Whitehorse batholith in the Yukon; and the 142 Ma Endako intrusions in central British Columbia. All of these collections indicate northward translations of between 1170 and 2290 km (Table 4). Discussion of these translation distances can be misleading if time is not taken into consideration; thus the calculated translation rates are used for comparison. It is assumed that northward motion of all terranes east of the Denali Fault (i.e., east of the Insular Belt; Fig. 1), relative to the craton, ended with accretion in Eocene time at about 50 Ma (e.g., 52 Ma, Bardoux and Irving 1989; 49.4 Ma, Symons and Wellings 1989; 50 Ma, Vandall and Palmer 1990; 51 Ma, Harris et al. 1998).

Most of the studies require average minimum translation rates of between 1.8 and 2.6 cm/a except for the Carmacks Group volcanic rocks, which require a much higher rate of 10.3 cm/a (Table 4; Fig. 10). The Mount Lorne stock yields a translation rate of 4.2 ± 1.4 cm/a, also greater than the average minimum rate for the majority of the collections (2.3 ± 0.3 cm/a) but still considerably less than the rate from the 5 Ma younger Carmacks Group volcanic rocks (Fig. 10). The translation rate of 4.2 cm/a for the Mount Lorne Stock suggests a ~50% coupling of the Stikine Terrane with the

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**Fig. 7.** Representative orthogonal step demagnetization plots. Squares are projected in the north–east–south–west plane and the triangles in the up (U) – north – down (D) – south plane.




Kula plate, which had an average northward tangential rate of ~8.0 cm/a with respect to the fixed North American craton (Kelley 1993). This amount of coupling is reasonable given the frictional losses expected for a terrane being "ground" along a continental margin in transpression. Conversely, the translation rate for the Carmacks Group requires 100% coupling.

The comparable translation distance yielded by the Mount Lorne stock to the other plutons that are up to 35 Ma older can be explained by examining the relative motions between the North American plate and the oceanic plates to the west. During the Early to Late Cretaceous (~130–83 Ma), the Farallon plate was in transition from a pre-130 Ma dominantly easterly–southeasterly motion to a post-95 Ma dominantly northward motion (Kelley 1993). Thus, relative to the craton, there would be only a minor relative poleward translation for this time interval, as shown between ~110 and 75 Ma (Fig. 10). Around ~83 Ma the Kula plate separated from the Pacific plate (Woods and Davies 1982; Kelley 1993) and its motion was dominantly northwards, highly oblique to, and faster than the motion of the North American craton. The majority of the northward translation of the Stikine Terrane, and ultimately of the terranes forming most of the Intermontane Belt, was during this later time interval, from 83 to ~50 Ma. Thus it is reasonable to suggest that a much faster rate between 75 and 50 Ma occurred, as indicated by the Mount Lorne result (Fig. 10).

The estimates for rotation are somewhat less internally consistent; however, most of the collections show a clockwise rotation of ~65 ± 11°, except for the Carmacks Group volcanic rocks and the considerably older Endako intrusions (Table 4). The Mount Lorne stock shows a clockwise rotation of 57 ± 11°, similar to those from the majority of the studies, indicating that most of the rotation of this part of the Stikine Terrane occurred after 75 Ma.

The results from the Mount Lorne stock are significantly different than those from the slightly younger 70 Ma Carmacks Group volcanic rocks that are exposed ~200 km to the northwest (Marquis and Globerman 1988; Johnston et al. 1996; Wynne et al. 1998) (Table 4; Fig. 1). The Mount Lorne stock predicts only about one half the translation distance and almost three times the amount of rotation compared with the Carmacks Group estimate (Table 4). The original data from the Carmacks Group gave a translation estimate of 13.4 ± 8.5° northwards and a clockwise rotation of 10.2 ± 20.7° (Marquis and Globerman 1988). Butler

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**Table 3.** Site-averaged paleomagnetic data for the Mount Lorne stock.

| Site | NT | NE | NR | *D* (°) | *I* (°) | $A_{95}$ (°) | *k* |
|---|---|---|---|---|---|---|---|
| ML01 | 11 | 4 | 3 | 8.5 | 76.4 | 9.4 | 42 |
| ML02 | 9 | 4 | 4 | 6.2 | 72.1 | 6.7 | 70 |
| ML03 | 12 | 4 | 7 | 210.4 | –78.2 | 3.4 | 187 |
| ML04 | 10 | 4 | 6 | 211.0 | –75.1 | 8.8 | 31 |
| ML05 | 12 | 3 | 8 | 215.6 | –72.4 | 4.7 | 97 |
| ML06 | 11 | 2 | 7 | 229.0 | –82.0 | 10.1 | 27 |
| ML07d | 13 | 12 | 0 | 9.0 | 72.9 | 4.6 | 88 |
| ML08 | 12 | 9 | 0 | 25.1 | 83.0 | 6.7 | 61 |
| ML09 | 13 | 8 | 1 | 8.4 | 75.9 | 5.9 | 76 |
| ML10 | 12 | 0 | 8 | 196.5 | –73.6 | 3.4 | 262 |
| ML11 | 10 | 0 | 0 | — | — | — | — |
| ML12 | 15 | 5 | 7 | 16.9 | 75.9 | 4.1 | 116 |
| ML13 | 10 | 4 | 5 | 207.3 | –80.0 | 7.8 | 45 |
| ML14 | 11 | 6 | 3 | 31.0 | 70.3 | 8.5 | 38 |
| ML15 | 14 | 8 | 4 | 224.6 | –69.5 | 5.1 | 75 |
| ML16 | 12 | 7 | 3 | 213.8 | –70.7 | 5.0 | 95 |
| ML17 | 13 | 2 | 7 | 194.3 | –78.1 | 8.1 | 41 |
| ML18 | 13 | 3 | 5 | 212.5 | –85.3 | 7.9 | 50 |
| ML19 | 12 | 7 | 5 | 205.0 | –73.0 | 4.6 | 91 |
| ML20 | 13 | 8 | 5 | 190.5 | –69.0 | 4.5 | 86 |
| Mean | 18 | | | 24.2 | 75.9 | 2.4 | 201 |

**Notes**: NT, total number of specimens; NE and NR, number of specimens measured and used in the averages calculated by end points or by remagnetization circles, respectively. The mean direction in geographic coordinates is given by its declination (*D*), inclination (*I*), radius of 95% confidence ($A_{95}$), and precision parameter (*k*) (Fisher 1953). Site ML07d is a mafic dike.

**Fig. 8.** Enlarged central segment of an equal-area stereonet showing all site means of the Mount Lorne stock (Table 3). Circles are granitoid sites, and the triangle is the mafic dike site. Cluster means are indicated by an X, with its cone of 95% confidence of Fisher (1953).

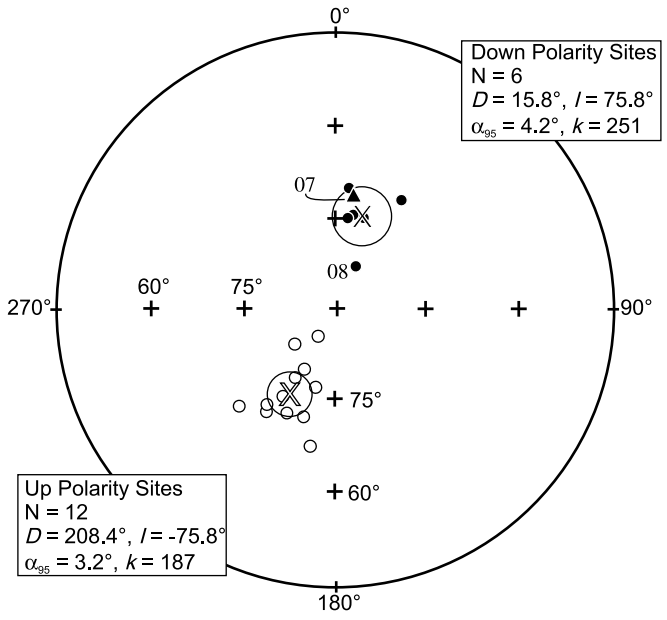



**Table 4**. Displacements of Cretaceous rock units within the Stikine Terrane.

| Unit | Location | Age (Ma) | Translation (°N)[a] | Translation (km)[a] | Average rate of motion (cm/a)[b] | Rotation (°)[a,c] | Reference |
|---|---|---|---|---|---|---|---|
| Carmacks volcanics | south Yukon | 70 | 22±6; 17.0±6.5 | 2420±660; 1900±700 | 11.1±3.0; 8.5±3.3 | 22±14CW; 4±14CW | Recalculated in this report; Wynne et al. 1998 |
| Mount Lorne stock | south Yukon | 75 | 10.5±3.5 | 1170±390 | 4.2±1.4 | 57±11CW | This study |
| Axelgold intrusions | central B.C. | 103 | 13.9±6 | 1540±670 | 2.6±1.1 | 55±13CW | Monger and Irving 1980 |
| Spences Bridge volcanics | south B.C. | 104 | 13.5±7 | 1500±780 | 2.5±1.3 | 66±13CW | Irving and Thorkelson 1990; Irving et al. 1995 |
| Mount McIntyre pluton | south Yukon | 109 | 14.3±5 | 1590±560 | 2.4±0.8 | 79±16CW | Harris et al. 1996 |
| Whitehorse batholith | south Yukon | 112 | 11.0±5 | 1220±560 | 1.8±0.8 | 59±17CW | Harris et al. 1997 |
| Endako intrusions | central B.C. | 142 | 20.2±11 | 2240±1220 | 2.2±1.2 | 3±17CCW | Symons 1983 |

[a]Calculation formulae are from Butler (1992, Appendix A), with reference poles interpolated from Besse and Courtillot (1991).
[b]Rates are calculated by dividing the translation distance by the unit age less the accretion age of 50 Ma (Bardoux and Irving 1989; Symons and Wellings 1989; Vandall and Palmer 1990; Harris et al. 1998).
[c]CW, clockwise; CCW, counterclockwise.

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**Fig. 9.** Locations of the Mount Lorne study area, its paleopole, and the ~75 Ma reference pole and APWP of Besse and Courtillot (1991).

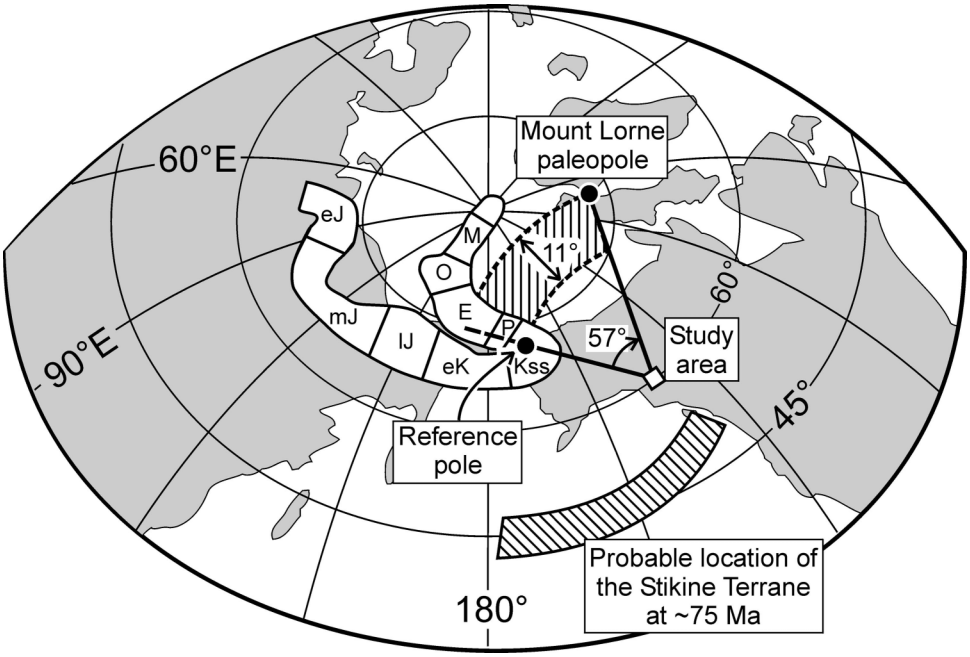
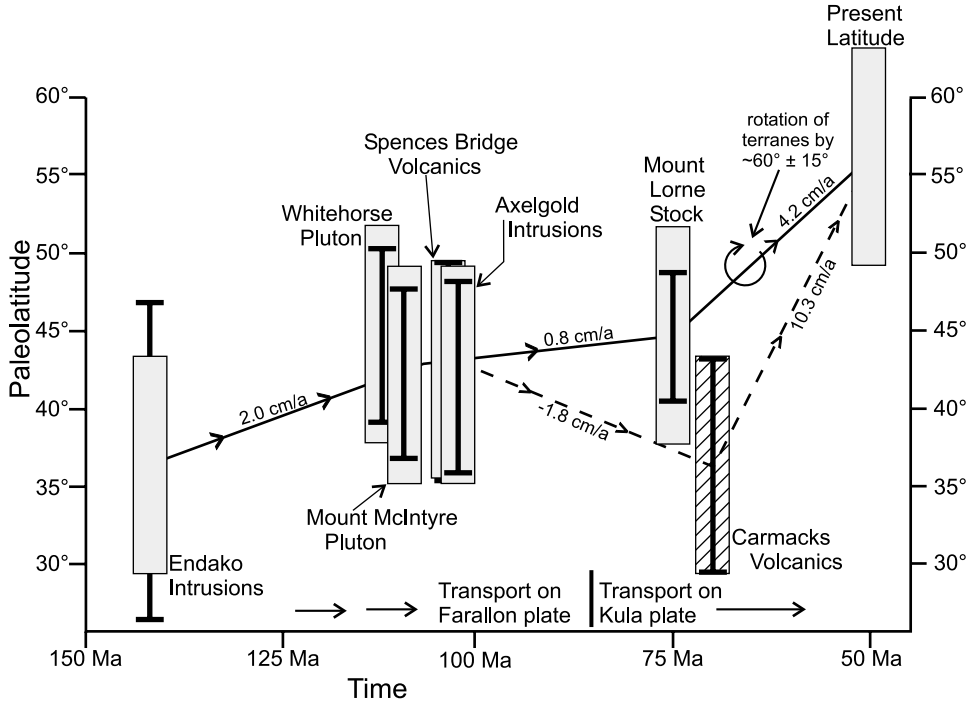



**Fig. 10.** Plot of paleolatitude for the Intermontane Belt versus time (after Harris et al. 1997). Blocks represent scaled dimensions of the present-day Intermontane Belt. Error bars on the blocks represent the 95% confidence errors (Table 4). Farallon and Kula plates at the bottom refer to the oceanic plate associated with the tectonic motion from Engebretson et al. (1985).




(1990) suggested that the original set of samples incompletely averaged out paleosecular variation. He also criticized the original tectonic estimate, deeming the 70 Ma reference pole to be unreliable. Johnston et al. (1996) revised the estimate, removing some of the original sites because they were deemed to be from an older volcanic suite, and added several new sites. Their result yielded a greater translation estimate of 17.2 ± 6.5° (1900 ± 700 km). Harris et al. (1996, 1997) noted a significant difference in the overall precision parameter for reversely polarized site mean ChRM directions compared with normally polarized site means, and a significant difference in directions between

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different sections of flows, indicating the possibility of either primary dips for the flows, incomplete averaging of paleosecular variation, or possibly insufficient magnetic cleaning. Most recently, Wynne et al. (1998) aimed to improve on the earlier estimates on the Carmacks Group and provide new temporal constraints on the paleogeography of the Cordillera. Thirteen new sites were added to the previously accepted 13 sites, although there was no significant change to the northward translation, the new estimate being 17.0 ± 6.5°. One concern with this result is the reference pole that Wynne et al. (1998) used. These authors averaged five poles for cratonic North America which they felt were reliable reference poles. We have used the accepted APWP of Besse and Courtillot (1991) because it is one of the more recently published paths for the North American craton, it is better constrained for the Cretaceous than previous paths, and the data that were used to determine the path had to meet a set criteria before being accepted. Thus, using the interpolated 70 Ma reference pole from the APWP of Besse and Courtillot (1991) of 192.7°E, 72.3°N ($A_{95}$ = 4.1), the recalculated Carmacks Group translation estimate increases from the published 17° (Wynne et al. 1998) to over 22° (Table 4), requiring an even greater translation rate.

In addition to the paleomagnetic and geotectonic problems with the Carmacks Group results, there are other concerns. The tectonic estimates led Johnston et al. (1996) and Wynne et al. (1998) to propose that the Carmacks Group volcanic rocks were extruded when their microplate overrode the Yellowstone hotspot at 70 Ma. This was supported by noting that the Carmacks lavas have a chemistry similar to that of the type absarokites of Wyoming (Johnston et al. 1996; Wynne et al. 1998). However, tenuous aspects of their hypothesis include the coeval widespread hydrothermal alteration of the lower Carmacks Group lavas which was noted by Johnston et al. (1996) and Wynne et al. (1998). The two studies causally link Carmacks volcanism with widespread remagnetization caused by hydrothermal activity. If the Carmacks Group is altered, then the hydrothermal event may postdate the extrusion at 70 Ma. This would suggest that the northward translation rate would be even greater than the excessive one already stated. Finally, there is minimal evidence for the existence of the Yellowstone hotspot at 70 Ma (cf. Lipman 1992; Anders 1994).

## Conclusions

Geobarometric analyses indicate that the Mount Lorne stock has not been significantly tilted since emplacement, but does suggest that there has been 1000 m of uplift on the eastern side of the stock compared with the western side along a normal fault. The paleomagnetic results suggest that the stock and the Stikine Terrane in the Yukon were translated northwards by 1170 ± 390 km and rotated clockwise by 57 ± 11° between ~75 Ma and 50 Ma. Excluding the Carmacks Group tectonic estimates, these results are statistically consistent with all other paleomagnetic studies in the Stikine Terrane. Further, the results support plate-tectonic models that suggest that the oceanic plates along the western margin of North America moved northward with an increased velocity during the Late Cretaceous and Early Tertiary (Engebretson et al. 1985; Umhoefer et al. 1989; Kelley 1993) (Table 4; Fig. 10).

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